

28/09/26
Tuesday

Data structures

Analytical problem solving -

Name: R. Risha

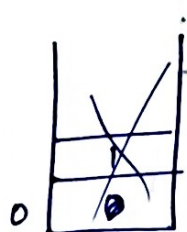
Reg no: 192525100

CSA0815

perform the following operations in LIFO data structure having an array size of 5. push(45), push(22), push(11), push(77), pop(), push(44), push(33), pop(), pop(), pop(). print peak element of the data structure and its index position. with a neat sketch explain in detail.

A.

Given an array of size 5.



Initially = $top = -1$
 $top = top - 1$

(i) push (45)



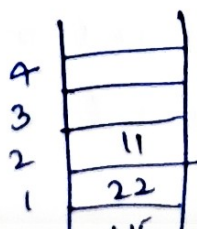
$top = top + 1$
 $= -1 + 1$
 $= 0$

(ii) push (22)



$top = top + 1$
 $= 0 + 1$
 $= 1$

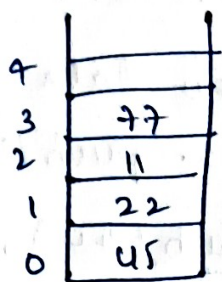
(iii) push (11)



$top = top + 1$
 $= 1 + 1$

28
36

(iv) push (77)



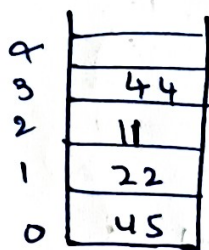
$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

(v) pop ()



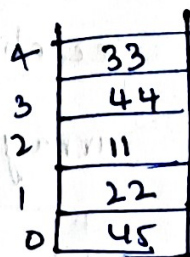
$$\begin{aligned} \text{top} &= \text{top} - 1 \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

(vi) push (44)



$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

(vii) push (33)



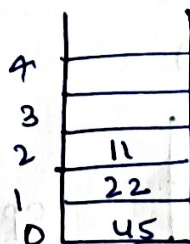
$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= 3 + 1 \\ &= 4 \end{aligned}$$

(viii) pop ()



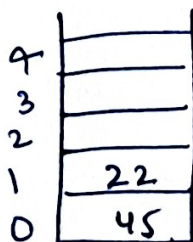
$$\begin{aligned} \text{top} &= \text{top} - 1 \\ &= 4 - 1 \\ &= 3 \end{aligned}$$

(ix) pop ()



$$\begin{aligned} \text{top} &= \text{top} - 1 \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

(x) pop ()



$$\begin{aligned} \text{top} &= \text{top} - 1 \\ &= 2 - 1 \\ &= 1 \end{aligned}$$

peak element = 22

22 is at position 1

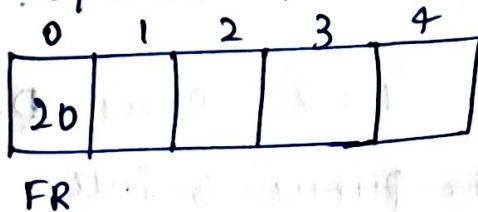
②

Perform the following operations in circular Queue having a size of 5. Enqueue (20), Enqueue (10), Enqueue (30), Dequeue (), Enqueue (80), dequeue (), Enqueue (90), Enqueue (100), dequeue (), dequeue (), Enqueue (60), Enqueue (90).

Given size = 5

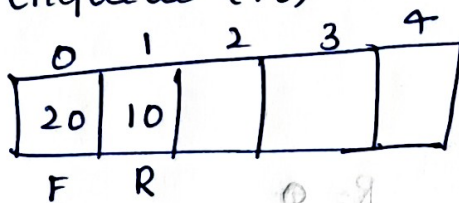
Initially $\boxed{\text{Front} = \text{Rear} = -1}$

(i) Enqueue (20)



$$F=0 \quad R=0$$

(ii) Enqueue (10)



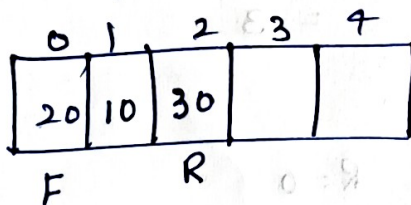
$$F=0$$

$$\text{Rear} = \text{Rear} + 1$$

$$= 0 + 1$$

$$= 1$$

(iii) Enqueue (30)



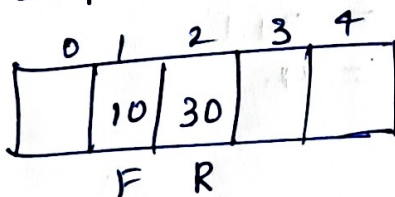
$$F=0$$

$$\text{Rear} = \text{Rear} + 1$$

$$= 1 + 1$$

$$= 2$$

(iv) Dequeue ()



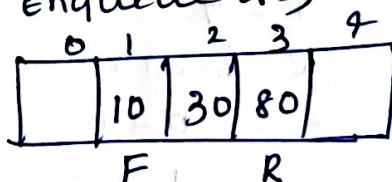
$$F = \text{Front} + 1$$

$$= 0 + 1$$

$$= 1$$

$$R = 2$$

(v) Enqueue (80)

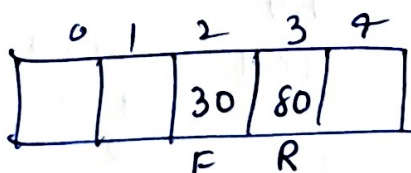


$$F=1 \quad R = \text{Rear} + 1$$

$$= 2 + 1$$

$$= 3$$

(vi) dequeue ()



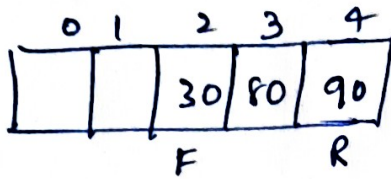
$$F = \text{Front} + 1$$

$$= 1 + 1$$

$$= 2$$

$$R = 3$$

(vii) Enqueue (90)



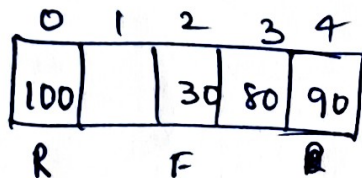
$$F=2$$

$$Rear = Rear + 1$$

$$= 3 + 1$$

$$= 4$$

(viii) Enqueue (100)



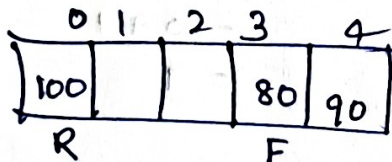
$$F=2$$

$$Rear = Rear + 1$$

$$\text{The Queue is Full} = -1 + 1$$

$$= 0$$

(ix) dequeue ()



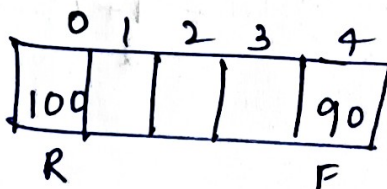
$$R = 0$$

$$F = Front + 1$$

$$(0 = 2 + 1)$$

$$= 3$$

(x) dequeue ()



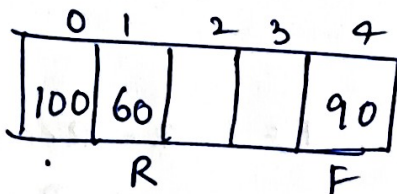
$$R=0$$

$$F = Front + 1$$

$$= 3 + 1$$

$$= 4$$

(xi) Enqueue (60)



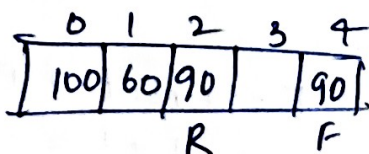
$$F=4$$

$$R = Rear + 1$$

$$= 0 + 1$$

$$= 1$$

(xii) Enqueue (90)



$$F=4$$

$$Rear = Rear + 1$$

$$= 1 + 1$$

$$= 2$$

So, $F=4$ $R=2$

Convert the following infix expression into a postfix expression.

$$B - (C/D + (E \% F * G) / H) * J.$$

Also, evaluate the given postfix expressions.

a) 9 3 8 4 / * + b) 6 2 + 6 3 / *

A.

Scanning
Character

Stack

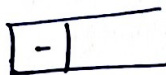
postfix
operation

B



B

-



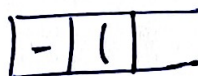
B (

(



B

c



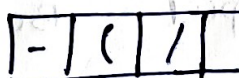
Bc

/



Bc

d



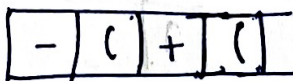
Bcd

+



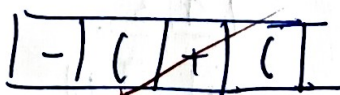
Bcd /

(



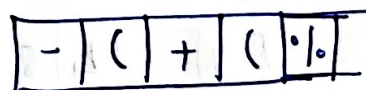
Bcd /

E



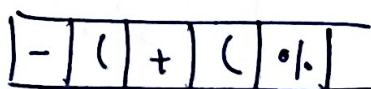
Bcd / E

%



Bcd / E

F



Bcd / EF

*



Bcd / EF %

9

-	1	+	(	*
---	---	---	---	---

Bcd / EF % G

)

-	(	+	(	*	)
---	---	---	---	--------------	--------------

Bcd / EF % G *

/

-	1	+	(	*	)	/
---	---	---	--------------	--------------	--------------	--------------

Bcd / EF % G *

H

-	(	+	/	/	
---	---	---	--------------	--------------	--

Bcd / EF % G * H

)

-			
---	--	--	--

Bcd / EF % G * H / +

wrong

*	
---	--

Bcd / EF % G * H / + -

J

*	
---	--

Bcd / EF % G * H / + - J

over

The output

Bcd / EF % G * H / + - J

a)

9 3 4 8 / * +

scanning key

stack

postfix operation

9

9

3

9	3
---	---

4

9	3	4
---	---	---

8

9	3	4	8
---	---	---	---

/

9	3	2
---	---	---

*

9	6
---	---

+

15

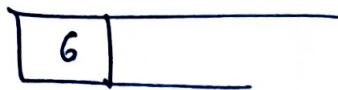
$$8/4 = 2$$

$$3 \times 2 = 6$$

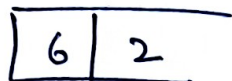
$$9 + 6 = 15$$

$$62 + 63 \mid *$$

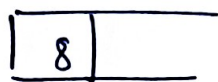
6



2



+

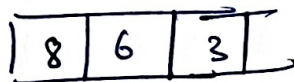


$$6 + 2 = 8$$

6

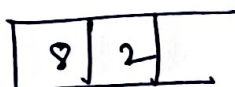


3

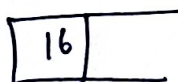


$$6/3 = 2$$

/



*



$$8 \times 2 = 16$$

output = 16